DevOps Documentation
Retail Sales and Performance Dashboard

Introduction:

The Retail Analytics System is designed to process and analyze sales data across multiple stores and products. This project leverages modern data engineering tools such as PySpark, MySQL, and MongoDB to build a scalable analytics pipeline. The system helps businesses monitor performance, identify trends, and make data-driven decisions. By integrating DevOps practices, the workflow is automated, ensuring faster and reliable data processing. The project simulates a real-world retail environment where large volumes of transactional data are generated daily and need to be analyzed efficiently.

Objective

The objective of this project is to analyze retail sales data to calculate revenue and profit, identify top-performing stores, and detect underperforming products for better business decision-making.

Technologies Used

Azure Databricks, PySpark, Delta Lake, Azure DevOps, YAML Pipelines, GitHub, Python, MySQL, and MongoDB.

ETL Architecture

Raw CSV Files → Bronze Layer → Silver Layer → Gold Layer → Analytics Reports

Data Processing

Bronze Layer stores raw data. Silver Layer performs cleaning and delayed shipment detection. Gold Layer generates supplier KPIs, inventory alerts, and delayed shipment reports.

DevOps Workflow

GitHub → Azure DevOps Pipeline → Databricks Notebook → Delta Tables → Reports & Logs

YAML Pipeline

```
trigger:

- main

pool:

  vmImage: 'ubuntu-latest'

steps:

- script: |pip install -r requirements.txt

  displayName: Install Dependencies

- script: |python python/preprocessing.py

  displayName: Run Preprocessing

- script: |python pyspark/retail_analysis.py

  displayName: Run PySpark Retail Analysis
```

Databricks Processing

```
df = df.withColumn("revenue", expr("quantity * sale_price")).withColumn("profit", expr("quantity * (sale_price - cost)"))

store_summary=df.groupBy("store_id").agg(sum("revenue").alias("total_revenue"),sum("profit").alias("total_profit"))

store_summary.write.format("delta").mode("overwrite").save("/tmp/store_summary")
```

Generated Reports

The system generates reports such as category-wise revenue, store performance, and top-selling products. These reports help businesses gain insights and make informed decisions based on data.

Benefits

The system automates data processing, reduces manual effort, and improves accuracy. It provides faster insights, supports scalability, and helps businesses make data-driven decisions efficiently.**11.**

Conclusion

This project demonstrates a complete retail analytics pipeline using modern tools and technologies. It highlights the importance of automation and scalable processing in generating valuable business insights from data.